

A Model hyperparameters

The hyperparameters for CLIPred and baselines are listed in Table 4. The hyperparameters for sequential objectives (I-JEPA $\rightarrow$ CLIP) is the same as I-JEPA only and CLIP-only during each objective’s training.

Table 4: Hyperparameters of CLIPred and baselines.

Hyperparameter	CLIP	IJEPA	CLIPred	DINOv2+CLIP
batch size (total)	400	400	400	400
epochs	300	1000	1000	1000
learning rate schedule	Cosine Decay	Cosine Decay	Cosine Decay	Cosine Decay
peak learning rate	0.001	0.001	0.001	2.0e-4
start learning rate	0.0002	0.0002	0.0002	0.0
final learning rate	1.0×10^{-6}	1.0×10^{-6}	1.0×10^{-6}	1.0×10^{-6}
warmup (in epochs)	10	40	40	80
weight decay	0.04	0.04	0.04	0.04
final weight decay	0.4	0.4	0.4	0.2
optimizer	AdamW [33]	AdamW	AdamW	AdamW
optimizer momentum	$\beta_1, \beta_2 = 0.9, 0.999$	$\beta_1, \beta_2 = 0.9, 0.999$	$\beta_1, \beta_2 = 0.9, 0.999$	$\beta_1, \beta_2 = 0.9, 0.999$
eps	1.0×10^{-8}	1.0×10^{-8}	1.0×10^{-8}	1.0×10^{-8}
start EMA momentum	/	0.996	0.996	0.994
final EMA momentum	/	1.0	1.0	1.0
ema schedule	/	Linear	Linear	Cosine
start teacher temp	/	/	/	0.04
final teacher temp	/	/	/	0.07
λ (CLIP loss weight)	/	/	0.01	0.1